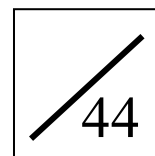


Pre S3 L04-06 Percentage Quiz
(ANSWER)

Name: _____

Date: _____



1. The original weight of a panda was 80 kg. Its weight first decreased by 10% due to illness and then increased by 10% after recovery. After recovery, what was the change in its weight as compared to the original weight?

- A. no change
B. an increase of 0.8 kg
C. a decrease of 0.8 kg
D. a decrease of 8 kg

Weight after recovery
 $= 80 \times (1 - 10\%) \times (1 + 10\%) \text{ kg}$
 $= 80 \times 0.9 \times 1.1 \text{ kg}$
 $= 79.2 \text{ kg}$
 Change in its weight
 $= (79.2 - 80) \text{ kg}$
 $= -0.8 \text{ kg}$
 $\therefore$ There was a decrease of 0.8 kg in its weight.

C

2. Last Christmas, an artist sold 40 postcards at the price of \$90 each for charity. This Christmas, the price of each postcard increases by 30% but the number of postcards sold decreases by 15%. Find the percentage change in the amount of money to charity.

- A. -15%
B. +15%
C. -10.5%
D. +10.5%

Last Christmas, money to charity = $\$90 \times 40$
 $= \$3\,600$
 This Christmas,
 price of each postcard = $\$90 \times (1 + 30\%)$
 $= \$117$
 number of postcards sold = $40 \times (1 - 15\%)$
 $= 34$
 amount of money to charity = $\$117 \times 34$
 $= \$3\,978$
 $\therefore$ Percentage change = $\$ \frac{3978 - 3600}{3600} \times 100\%$
 $= \$ \frac{378}{3600} \times 100\%$
 $= +10.5\%$

D

3. Bob deposits 50000 in bank A at a simple interest rate of 6% p.a. and \$40000 in bank B at a simple interest rate of 7% p.a. Find the total amount he will receive after 10 years.

- A. \$148000
B. \$74000
C. \$58000
D. \$12000

Amount received from bank A
 $= \$50\,000 \times (1 + 6\% \times 10)$
 $= \$80\,000$
 Amount received from bank B
 $= \$40\,000 \times (1 + 7\% \times 10)$
 $= \$68\,000$
 Total amount he will receive
 $= \$ (80\,000 + 68\,000)$
 $= \$148\,000$

A

4. A sum of money is deposited in a bank at an interest rate of 12% p.a. compounded yearly. If the interest received after 7 years is \$6000, find the principal.
(Give the answer correct to the nearest \$1000.)

- A. \$3000
B. \$4000
C. \$5000
D. \$6000

Let \$P be the principal.
 $P(1 + 12\%)^7 - P = 6\,000$
 $1.12^7 P - P = 6\,000$
 $(1.12^7 - 1)P = 6\,000$
 $P = \frac{6\,000}{1.12^7 - 1}$
 $= 5\,000, \text{ cor. to the nearest } 1\,000$
 $\therefore$ The principal is \$5 000.

C

5. In this financial year, Tom has a total allowance of \$140000 and he has to pay a salaries tax of \$15300. If his net chargeable income is greater than \$150000 but less than \$200000, find his annual income.

- A. \$109000
B. \$195000
C. \$249000
D. \$335000

Tom's salaries tax payable
 $= \$ (50\,000 \times 2\% + 50\,000 \times 6\% + 50\,000 \times 10\% + x \times 14\%)$
 $= \$ (9\,000 + 0.14x)$
 $\therefore 9\,000 + 0.14x = 15\,300$
 $0.14x = 6\,300$
 $x = 45\,000$
 Tom's net chargeable income = $\$ (150\,000 + 45\,000)$
 $= \$195\,000$
 Tom's annual income = $\$ (195\,000 + 140\,000)$
 $= \$335\,000$

Net chargeable income	Rate
On the first \$50 000	2%
On the next \$50 000	6%
On the next \$50 000	10%
On the next \$50 000	14%
Remainder	17%

D

Pre S3 L04-06 Percentage Quiz
(ANSWER)

6. The value of an antique oil painting increases at a steady rate of 25% every 5 years. Its present value is \$150000.

(a) Find its value 20 years ago. (4 marks)

(b) Find the increase in its value over the past 20 years. (2 marks)

(a) Let \$P\$ be the value of the painting 20 years ago.
 Taking every 5 years as a period,
 number of periods in 20 years = $\frac{20}{5} = 4$ 1A
 $150\,000 = P(1 + 25\%)^4$ 1M+1A
 $P = \frac{150\,000}{1.25^4}$
 $= 61\,440$
 $\therefore$ The value of the painting 20 years ago was \$61 440. 1A

(b) Increase in value
 $= \$ (150\,000 - 61\,440)$ 1M
 $= \$88\,560$ 1A

7. David borrows \$5000 from a bank at an interest rate of 7.8% p.a. compounded monthly. Find the amount he should repay after 3 years.

(Give the answer correct to the nearest dollar.) (4 marks)

Interest rate per month = $\frac{7.8\%}{12} = 0.65\%$ 1A
 Taking a month as a period,
 number of periods in 3 years = $3 \times 12 = 36$ 1A
 Amount = $\$5\,000 \times (1 + 0.65\%)^{36}$ 1M
 $= \$6\,313$, cor. to the nearest dollar 1A

- *8. Bank X offers interest at 4% p.a. compounded half-yearly. Bank Y offers interest at 2% p.a. compounded quarterly. Susan is going to deposit a sum of money in one of the banks. She claims the interest received in bank X will be more than that in bank Y after 10 years. Do you agree? Explain your answer. (10 marks)

Let \$P\$ be the sum of money deposited.
 For bank X:
 interest rate for half a year = $\frac{4\%}{2} = 2\%$ 1A
 Taking half a year as a period,
 number of periods in 10 years = $10 \times 2 = 20$ 1A
 Interest received after 10 years = $\$[P(1 + 2\%)^{20} - P]$ 1M+1A
 $= \$ (1.02^{20} - 1)P$ 1A
 For bank Y:
 interest rate per quarter = $\frac{2\%}{4} = 0.5\%$ 1A
 Taking a quarter as a period,
 number of periods in 10 years = $10 \times 4 = 40$ 1A
 Interest received after 10 years = $\$[P(1 + 0.5\%)^{40} - P]$ 1A
 $= \$ (1.005^{40} - 1)P$ 1A
 $\therefore 1.02^{20} - 1 = 0.486$, cor. to 3 d.p.
 $1.005^{40} - 1 = 0.221$, cor. to 3 d.p.
 $\therefore \$ (1.02^{20} - 1)P > \$ (1.005^{40} - 1)P$ 1M
 $\therefore$ The claim is agreed. 1A